

Type of self-talk matters: Its effects on perceived exertion, cardiorespiratory, and cortisol responses during an iso-metabolic endurance exercise

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Abstract

Self-talk is an effective mental training technique that has been shown to facilitate or debilitate an athlete's performance, depending on its valence. Although the effects of self-talk have been supported by observing change in sport performance, little is known about how self-talk can induce physiological changes. Specifically, it is important to understand if the type of self-talk (positive, neutral, or negative) and can influence stress-related parameters, such as perceived exertion, cardiorespiratory, and cortisol responses. The study's objective was therefore to investigate the top-down effect of positive and negative self-talk compared to a dissociative activity during an iso-metabolic running exercise on autonomic regulation of cardiorespiratory function. Twenty-nine well-trained male runners [38 ± 13 years, 177 ± 7 cm and 73 ± 7 kg] volunteered to participate in a randomized-group design study that included a negative self-talk (NST), a positive self-talk, and a dissociative group (DG). First, participants underwent an incremental running test on a treadmill to determine the maximal oxygen uptake ($\dot{V}O_{2max}$). Next, participants received a mental training session on self-talk and created three positive and three negative self-talk statements. Finally, participants underwent a 60-min steady-state running exercise on a treadmill at 70% of $\dot{V}O_{2max}$, during which they were cued at 20-, 35-, and 50-min with their personal self-created positive or negative self-talk statements while the DG listened to a documentary. Cardiorespiratory parameters and rate of perceived exertion (RPE) were recorded throughout the 60-min endurance exercise. In addition, salivary cortisol samples were obtained at waking and after treatment. Although oxygen uptake, carbon dioxide production, RPE, and heart rate significantly changed overtime during the 60-min steady-state running exercise, no significant main treatment effect was found. However, RPE scores, minute ventilation, breathing frequency, and salivary cortisol were significantly higher in the NST group compared to the two other groups. These data suggest that NST [emotion-induced stress, as reflected by elevated cortisol] altered the breathing frequency response. In conclusion, manipulating self-talk alters hormonal response patterns, modulates cardiorespiratory function, and influences perceived exertion.

KEYWORDS

cardiorespiratory response, cortisol, endurance running exercise, perceived exertion, self-talk

1 | INTRODUCTION

Anyone who performed prolonged physical exercise has experienced the inner conversation that encourages them to keep going or challenges their motives to cope with increasing discomfort (Gammage et al., 2001). Self-talk refers to a syntactically recognizable articulation of an internal position that can be expressed internally or out loud where the sender of the message is also the receiver (Conroy & Coatsworth, 2007; Raalte & Vincent, 2017). During a physical effort, self-talk arises from a conscious perception of emotional and physiological cues relevant enough to create awareness about exercise-induced stress (St Clair Gibson & Foster, 2007).

Self-talk, a mental construct, contains various conceptual elements (Raalte & Vincent, 2017) classified into five themes: nature, structure, person, task instructions, and miscellaneous (Hardy, 2001). The nature dimension refers to positive and negative self-talk (e.g., “I can do this” and “No way I can do this,” respectively), while structure concerns single words, phrases, or complete sentences; the person dimension applies to self-talk said in the first or second person, the task instruction ascribes categories as skill-specific (e.g., “Shoot the ball”) or general (e.g., “Go faster”) and miscellaneous may include thoughts about work, career, personal problem-solving or unintelligible inner chatter (Aitchison et al., 2013; Hardy, 2001; Hardy et al., 2005).

Previous studies (St. Clair Gibson et al., 2003; St Clair Gibson & Foster, 2007) posited a relationship between the conscious awareness of exercise stress, the internal focus (i.e., association), and the exercise intensity. Accordingly, associative self-talk becomes more prevalent with the increment of exercise intensity or duration and the increased rate of perceived exertion (RPE). It includes thoughts about feelings and affects body monitoring, pain, command, instruction, and pace monitoring (Hardy, 2001; Johnson & Siegel, 1992; Masters & Ogles, 1998; St Clair Gibson & Foster, 2007). Associative thoughts occur during endurance exercises above 70% of maximal running speed (MAS) with RPE scores within 16–20 using the 6–20 Borg’s scale (Aitchison et al., 2013; Borg, 1982; Schomer & Connolly, 2002). Dissociation (i.e., externalized attentional focus), on the contrary, directs attention away from the exercise stress and the peripheral physiological changes (Masters & Ogles, 1998; St Clair Gibson & Foster, 2007), a mental state that spontaneously prevails in low-intensity exercise within RPE scores of 6–10 (Aitchison et al., 2013; St Clair Gibson & Foster, 2007). Noteworthy,

during moderate exercise equivalent to 70% of MAS (RPE 11–15), Aitchison et al. (2013) showed no difference in the amount of associative and dissociative thoughts among runners.

Association, however, may be positive or negative in nature. For instance, when integrating pain sensations during exercise, one may produce negative thoughts reinforcing that pain is unbearable or instead, use pain as motivational self-talk to cope with exercise stress to increase or maintain performance. In that regard, Hatzigeorgiadis et al. (2011) conducted a meta-analysis to feature the up-to-date outcomes on this topic. They reported a positive moderate effect size ($ES = .48$) of self-talk interventions on task performance in sport. In the same year, Tod et al. (2011) published a systematic review that highlighted the beneficial effects of positive self-talk on performance while reporting no impediment of performance due to negative self-talk. However, certain athletes interpret the negative self-talk as a challenge to which they positively respond to improve exercise performance, while others consider it anxiety-producing and counterproductive (Hamilton et al., 2007). Regarding the latter, the interpretation of negative associative thoughts during exercise may increase the mental effort in worry or anxiety and cause an increase in on task effort and trigger the response to stress mediated by the hypothalamus-pituitary-adrenal axis (HPA) activity (Verkuil et al., 2010; Wilson et al., 2007). Besides, strategies to induce dissociation by way of music or video may reduce RPE, increase time-to-exhaustion and even aid in dimming down negative body sensations during high-intensity exercise (Chow & Etnier, 2017; Maddigan et al., 2019).

Although the content and interpretations of self-talk vary among athletes, its general purpose is to regulate exercise and enhance performance (Hardy, 2001; Hardy et al., 2005). It means that the decision to increase and decrease velocity (i.e., self-paced) or tolerate or terminate the exercise session (i.e., exhaustion) is controlled by the prefrontal cortex (PFC) through a bi-directional mind/body integration (Robertson & Marino, 2020). That is, top-down neural connections, initiated through declarative or non-declarative mental processing in the PFC, regulate motor cortex outputs and muscle recruitment. Meanwhile, somatosensory feedback modulates the central neural drive via the bottom-up afference to the brainstem, limbic system, and cerebral cortex (Bechara et al., 2000; Damasio, 1996; Taylor et al., 2010; Thayer & Lane, 2000). Within the bi-directional brain/body framework, the interpretation of associative motivational self-talk would increase the top-down control of action,

physical effort, and the sympathetic drive (Bellomo et al., 2020; Hatzigeorgiadis et al., 2011) as physiological changes due to the exercise stress convey information from the periphery to the central nervous system affecting RPE, the self-talk interpretation, cardiorespiratory response and self-pace (St Clair Gibson & Foster, 2007; St Clair Gibson et al., 2006; Williamson, 2010). In addition, induced dissociation can alter the interplay between central motor drive, central cardiovascular command, and perceived exertion due to the limited PFC capacity to process both the peripheral feedback and the induced external stimulus during moderate-to-high exercise intensities (Fontes et al., 2020; Maddigan et al., 2019; Rejeski, 1985).

Previous studies have investigated the interplay between self-talk and RPE during self-paced time trials (Aitchison et al., 2013; Baden et al., 2004; Blanchfield et al., 2014; Schomer & Connolly, 2002; St Clair Gibson & Foster, 2007). Self-pace, however, is modulated through the continuous processing of feedforward and feedback information (Baden et al., 2004; St Clair Gibson et al., 2006). In this context, the direction and weight of cause-and-effect events are difficult to unravel. Accordingly, the current study applied a design intended to match the relative exercise workload (iso-metabolic/physiological stress) among experimental groups to control for the somatosensory feedback and enhance the top-down effect of the induced associative and dissociative self-talk on the exercise stress response. The study's objective was to investigate the effect of associative positive and negative self-talk compared to an induced dissociative activity during a one-hour steady-state running exercise at 70% MAS. The exercise intensity was chosen as a "gray intensity zone" because of its random prevalence of associative and dissociative thoughts that could potentially match the perceived exertion (Aitchison et al., 2013). Therefore, the study was conducted to determine whether the associative negative/positive self-talk during a one-hour steady-state running exercise would impact the acute cardiorespiratory response, alter the perception of exertion and the HPA

activity compared to the dissociative activity. Theoretical and experimental perspectives are discussed based on the exploratory analysis of the outcomes.

2 | METHOD

2.1 | Subjects

Twenty-nine well-trained male runners volunteered to participate in the study. Participants were recruited from a University Cross-Country running team and local running clubs. The mean age, height, and mass were 38 ± 13 years, 176 ± 6 cm, and 73 ± 5 kg; 40 ± 13 years, 178 ± 7 cm, and 76 ± 10 kg; and 36 ± 13 years, 178 ± 8 cm, and 69 ± 6 kg for the negative self-talk group (NST), positive self-talk group (PST), and dissociative group (DG), respectively. All participants were injury-free and motivated to perform during the tests. Participants' characteristics and training profiles are shown in Table 1. Participants were fully informed of the study procedures and provided consent before participating in accordance with Memorial University's Human Investigation Committee (HIC) regulations. All experiments were carried out under the Declaration of Helsinki.

The participants attended six sessions over a maximum of fourteen days (See Figure 1). During session one, participants were familiarized with the testing procedures, read and signed the Consent Form, and filled in the Physical Activity Readiness Questionnaire (Par-Q). The anthropometric measurements were also recorded.

During session two, the participants performed an incremental running test on the treadmill to determine their maximal oxygen uptake ($\dot{V}O_{2max}$) and its associated cardiorespiratory parameters along with the maximal aerobic speed (MAS). Before testing, participants were instructed to (1) fast for at least 4 hr, (2) abstain from strenuous exercise for 24 hr, and (3) restrain from caffeine and alcohol intake, as well as tobacco inhalation. They were also screened for the following conditions: prescribed

TABLE 1 Athletes' characteristics and training profile

	Maximal aerobic speed (km hr ⁻¹)	Maximal heart rate (beat min ⁻¹)	Training experience (year)	Training session (Nb week ⁻¹)	Interval training (units >75%VO _{2max})	Training load (hr:min week ⁻¹)	10 km personal best (min:sec)
Negative self-talk	16.6 ± 1.6	178 ± 11	7.9 ± 9.8	4.9 ± 1.0	2.8 ± 0.5	5:20 ± 1:20	38:30 ± 3:20
Positive self-talk	17.3 ± 1.6	187 ± 9	7.2 ± 4.9	5.2 ± 1.3	2.4 ± 0.8	6:06 ± 2:11	38:20 ± 3:29
Dissociative group	17.3 ± 1.1	184 ± 6	8.5 ± 8.5	5.2 ± 0.9	2.1 ± 1.1	5:48 ± 1:36	36:31 ± 2:09
All groups average	17.1 ± 1.4	183 ± 9	7.8 ± 7.7	5.1 ± 1.1	2.4 ± 0.8	5:42 ± 1:42	37:20 ± 2:55

Note: Mean ± SD.

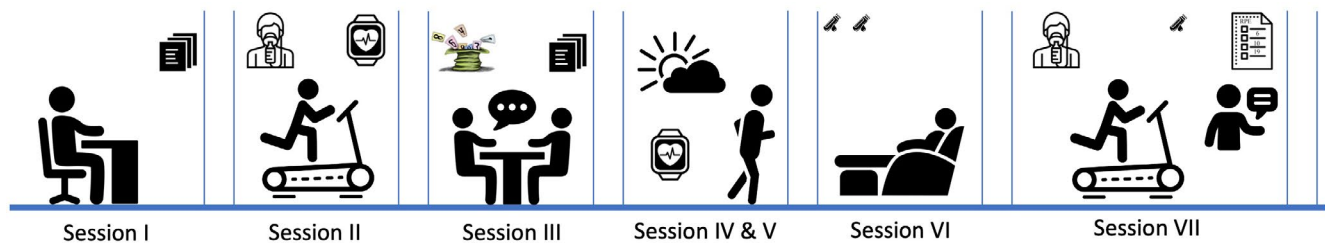


FIGURE 1 Timeline of the study over a maximum of fourteen days. In session I, participants were familiarized with the testing procedures and completed the Consent Form and Physical Activity Readiness Questionnaire (Par-Q). In session II, participants underwent an incremental running test on the treadmill. In session III, participants were randomly assigned to a group and received a mental training session on self-talk. In session IV and V, participants ran outdoor for one-hour at 70% maximal aerobic speed (MAS) to practice self-talk. Session VI, participants had a rest day during which baseline cortisol was measured. In session VII, participants partook in a steady-state running exercise at 70% MAS

medication, exercise-induced fatigue, blood pressure above 140 mmHg, and injury. The presence of screened conditions led to testing postponement or dismissal from the study.

During session three, participants were randomly assigned to NST ($n = 10$), PST ($n = 9$), or DG ($n = 10$) and received a mental training session on self-talk delivered by the investigator. The session overviewed research findings on how the psychological effects of positive and negative self-talk can affect performance across various sports, including running. In addition, this session discussed how the magnitude effect on performance could depend on the intensity of the self-talk (e.g., slightly encouraging/discouraging, moderately encouraging/discouraging, and highly encouraging/discouraging). At the end of the 40-min session, the participants were requested to create their self-talk statements in three categories: (i) slightly encouraging self-talk (level 1), (ii) moderately encouraging self-talk (level 2), and (iii) highly encouraging self-talk (level 3). To create the “slightly encouraging self-talk”, participants were asked to write helpful statements that would give them a slight boost at the start of a race. To create “moderately encouraging self-talk”, participants were asked to write helpful statements they would use half-way through a race when feeling moderately exhausted. Finally, to create “highly encouraging self-talk”, the participants were asked to create helpful statements they would use near the end of the race to cope with exercise-induced fatigue. The participants then repeated this process by creating negative self-talk statements in three parallel categories: (i) “slightly negative self-talk”, (ii) “moderately negative self-talk”, and (iii) “highly discouraging self-talk” statements (see Supporting Information for examples). These statements were later cued to participants during the steady-state running exercise at 70% MAS.

During sessions four and five, all participants were requested to run for one-hour at 70% MAS determined from the participant's incremental test results and monitored by

a GPS-enabled sports watch (Model Forerunner 205/305, Garmin Ltd, Kansas City, TX) along with practicing positive self-talk statements. Participants assigned to the negative group were also asked to practice associative thoughts with positive self-talk statements because it represents a state of mind coherent with moderate steady-state running exercise within RPE scores of 11–15 with no pressure for performance enhancement (Aitchison et al., 2013). The aim was to practice associative self-talk *per se* during a mild exercise intensity and volume performed by experienced endurance runners. In this scenario, associative and dissociative self-talk flows in random prevalence (Aitchison et al., 2013). The use of this task ensured that all participants had the experience of consciously inducing the associative self-talk during mild intensity running that was dissimilar to the experimental lab task (i.e., treadmill vs. overground running). Thus, the purpose was to practice the associative self-talk to enhance the technique's effectiveness (Hatzigeorgiadis et al., 2011) during a well-controlled treadmill exercise.

After a 24-hr rest, participants partook, during the sixth session, in a steady-state running exercise at 70% MAS at the Human Physiology Laboratory. Prior to attending the session, they were reminded to comply with the instructions provided during the second session. Any disregard of the criteria led to testing postponement or dismissal from the study. After each running test, the participants could cool down until the displayed heart rate indicated a reading of 100 BPM or lower.

2.2 | Testing protocol

2.2.1 | Maximal oxygen uptake determination protocol

The incremental test was performed on a motor-driven treadmill at a constant 1% slope (Trackmaster, modified model TMX55, JAS Fitness Systems, Newton, KS). After

a 5-min warm-up at a speed of 5 km hr⁻¹, the initial speed was set at 7 km hr⁻¹, and afterward increased by 1 km hr⁻¹ every two minutes until voluntary exhaustion (Leger & Boucher, 1980). The incremental test was followed by a supra-maximal square-wave running test (verification phase) at 105% MAS after a 5-min recovery period to confirm a true $\dot{V}O_{2max}$ (Rossiter et al., 2006). A higher $\dot{V}O_2$ value during the verification phase invalidated the incremental running test. From the post-acquisition analysis and according to previous research (Basset & Boulay, 2003), the running velocity at the final stage (MAS) was determined as follows: (i) the 2-min stages were divided into four quarters corresponding to 0.25, 0.50, 0.75, and 1 km hr⁻¹; (ii) the $\dot{V}O_{2max}$ was determined as the highest thirty-second average during the running test; and (iii) the time corresponding to this value was matched with the corresponding quarter of the 2-min stage. For instance, a runner who reached $\dot{V}O_{2max}$ at 100-s into the 17 km hr⁻¹, had a 0.75 km hr⁻¹ added to the actual velocity to bring his MAS up to 17.75 km hr⁻¹.

2.2.2 | Steady-state running exercise

Participants ran for an hour at 70% MAS with a constant 1% slope on the same motor-driven treadmill as the incremental test. Every 5-min, the participants were asked to estimate their perceived exertion on the Borg 6–20 RPE scale. The exercise intensity of 70% MAS was selected because previous findings have shown that exercise at an RPE score above 15 results in a shift from dissociative to associative thoughts (Schomer & Connolly, 2002); moreover, within an RPE score range of 11–15, there was no difference in the prevalence of associative or dissociative thoughts (Aitchison et al., 2013). In the present study, the average RPE score increased from 11 (at 5-min) to 14 (at 60-min) during the steady-state running exercise at 70% MAS. Therefore, the paradigm consisted of reducing cognitive conflict with the physiological effort, both by giving cues to participants for assisted associative self-talk or by inducing dissociation.

The dissociative group (DG) was assigned the task of listening to a documentary titled: “Steven Hawking: Master of the Universe”, on a Sony MP3 player (model NWZ-E436F) while they performed their run. The science-fiction documentary was used to distract the participant from invoking mental training techniques that could potentially enhance performance. Participants were asked to pay attention to the documentary and advised that they may be asked questions about the content. In addition, the participants were instructed to place the MP3 player in a position with which they felt most comfortable, either attached to the treadmill ramp or on an arm strap. Thus, the

effects of the nature of self-talk (i.e., PST and NST groups) were compared to a dissociative running task with absence, or at least low prevalence, of associative thoughts of any nature. In other words, the top-down effect of spurious associative thoughts may have been attenuated due to the external distraction.

Individuals slight, moderate, and high NST or PST statements were cued starting at the 20-, 35-, and 50-min marks, respectively. Participants' slight and moderate self-talk statements were verbally cued five times in a row every 5-min between minutes 20–30 (i.e., 5-times at minutes 20, 25, and 30) and 35–45 (i.e., 5-times at minutes 35, 40, and 45), respectively. At the 50-min mark, high self-talk statements were verbally cued every minute. For all statements, the participants were required to repeat them aloud for a total of forty cue-and-repeat pairs. Progressively cuing in different positive and negative levels of self-talk was a novel and exploratory approach. This approach attempted to keep the participants focused and attentive on the specific statements designed for the PST and NST groups, avoid desensitization, and minimize the opportunity to avail of alternative coping strategies.

2.2.3 | Cardiorespiratory measurements

During both the incremental test and steady-state running exercise, oxygen uptake ($\dot{V}O_2$), carbon dioxide production ($\dot{V}CO_2$), breathing frequency (Bf), and tidal volume (V_T) were continuously collected with an indirect calorimetry system implemented with O_2 and CO_2 analyzers (Model S-3A and Anarad AR-400, Ametek, Pittsburgh, PA), and with a pneumo-tachometer (Model S-430, Vacumetrics/Vacumed Ltd., Ventura, CA) connected to a 4.2 L mixing chamber. Respiratory exchange ratio (RER) and minute ventilation ($\dot{V}_E$) were calculated as the quotient of $\dot{V}CO_2$ on $\dot{V}O_2$ and as the product of Bf by V_T , respectively. In addition, HR data were wirelessly transmitted via telemetry to the AEI indirect calorimetric system with a Polar HR monitor (Polar Electro, Oy, Finland). Before testing, volume and gas analyzers were calibrated with a 3 L calibration syringe and medically certified O_2 and CO_2 calibration gases of 16% O_2 and 4% CO_2 , respectively. All calibrations were performed at the same location in a thermo-neutral environment. The data were online digitalized from an A/D card to a computer for monitoring the metabolic rate.

2.2.4 | Salivary cortisol

Salivary cortisol was measured as a marker of physiological response to acute stress (Basset et al., 2006; Hayes et al., 2016). Samples were obtained immediately after

waking and 30-min later on the resting day to minimize the potentially anticipatory effect of the forthcoming exercise bout and immediately post-intervention. For the sake of measurement consistency, the two first samples (delta) were analyzed to detect any undue stress prior to the intervention. Samples were collected by stimulating saliva flow by chewing on a salivette (IBL, Hamburg, Germany) for 1-min. Soaked salivettes were carefully placed in an aseptic and airtight tube and stored at -20°C prior to further analysis. After thawing, salivettes were centrifuged at 2,000–3,000 rpm for 5-min and 100 μl of the recovered supernatant was used for duplicate analysis employing a time-resolved immunoassay with fluorescence detection (Medicor Inc, Montréal, Qc) as previously described (Dressendorfer et al., 1992).

2.3 | Data reduction and analyses

The breath-by-breath metabolic data, along with the 5-sec mean heart rate, were averaged into 1-min blocks for incremental test and 5-min blocks for steady-state running exercise, which served as a data smoothing technique using Igor Pro 6.2 (Wave Metrics, Lake Oswego, OR). All data collected were aligned with respect to time and rate of perceived exertion. The RPE and cortisol scores were normalized from baseline values—first 5-min and first sample for RPE and cortisol, respectively—to account for inter-individual variations. Scores were then expressed as delta values.

2.4 | Statistical analysis

All variables are presented as mean ($\pm\text{SD}$). Levene test for equality of variances has been performed, and if significant, logarithmic adjustments were made. One-way analysis of variance (3 groups) was performed on anthropometric characteristics, training status, maximal cardiorespiratory parameters, and running performance. In addition, a two-way analysis of variance [3 groups (DG – NST – PST) $\times$ 12 segments of time (from 5 to 60-min)] was computed for all cardiorespiratory parameters. After log transformation, a two-way analysis of variance [3 groups (DG – NST – PST) $\times$ 3 epochs (waking, waking + 30 min, and post-treatments)] for repeated measures was computed for salivary cortisol. The 6-to-20 Borg scale is an ordinal scale that does not meet the assumption of normality (Vincent & Weir, 1994). Therefore, a Kruskal–Wallis test was used to detect any significant RPE score change through time and between groups. Statistical significance was set at $p < .05$. Statistical Package for the Social Sciences 19.0 was used for all statistical procedures (SPSS inc., Chicago, IL).

3 | RESULTS

3.1 | Participants' anthropometrics and training profile

Table 1 displays participants' characteristics, training profile, and running performance. No significant difference was detected between groups on anthropometrics, training parameters, or physical performance. Although non-significant, DG performed better on 10 km road race, being around 2-min faster than the two other groups. On the Mercier scoring table [<http://myweb.lmu.edu/jmureika/track/mercier/Merc99.html>], the runners ranked on average from 344, 350 to 426 points for NST, PST, and DG, respectively. These scores ranging from 34 to 43 percentile of the 10,000 m world record confirmed that the participants represented a good cluster of well-trained runners.

3.2 | Maximal cardiorespiratory variables—The incremental test

Table 2 displays participants' scores on the incremental test. No significant interaction or main group effect was detected on $\dot{V}\text{O}_2$ (absolute and relative), $\dot{V}\text{CO}_2$, $\dot{V}_E$, Bf, RER, and HR. A verification phase was conducted at the end of the incremental test to ensure a true $\dot{V}\text{O}_{2\text{max}}$ and most of the participants ($n = 24$) completed it. Twenty-one of the participants reached a higher $\dot{V}\text{O}_2$ value on the incremental test compared to the verification phase. Although non-significant the differences were 368.89 ± 72.51 , 235.62 ± 65.62 , and $686.52 \pm 110.44 \text{ ml min}^{-1}$ for NST ($n = 10$), PST ($n = 7$) and DG ($n = 7$), respectively. Two participants underscored the incremental test by an irrelevant amount of O_2 (-50 ml min^{-1}) compared to the verification phase, and one underperformed by $-145.40 \text{ ml min}^{-1}$. The five remaining individuals felt too exhausted at the end of the incremental test to undergo an additional time-to-exhaustion test. In these instances, the following criteria were applied for the determination of a true $\dot{V}\text{O}_{2\text{max}}$: a plateauing of oxygen uptake (an increase of less than $0.5 \text{ ml min}^{-1} \text{ kg}^{-1}$) despite an increase in speed, an RER of about 1.1, and a respiratory oxygen equivalent of 35 and above. These outcomes confirmed the homogeneity among participants in terms of cardiorespiratory functions.

3.3 | Steady-state running exercise

Figure 2 displays $\dot{V}\text{O}_2$, $\dot{V}\text{CO}_2$, RER, and HR over the 60-min steady-state running exercise. On average, participants ran at a constant speed of $12 \pm 0.3 \text{ km hr}^{-1}$ on the ergometer. Although three out of four parameters significantly

TABLE 2 Maximal cardiorespiratory parameters

	Absolute oxygen uptake (ml min ⁻¹)	Relative oxygen uptake (ml min ⁻¹ kg ⁻¹)	Carbon dioxide output (ml min ⁻¹)	Minute ventilation (L min ⁻¹)	Breathing frequency (breath min ⁻¹)	Respiratory exchange ratio (AU)
Negative self-talk	4,277 ± 524	56.7 ± 6.9	4,336 ± 566	184 ± 43	68 ± 11	1.01 ± 0.4
Positive self-talk	4,244 ± 591	59.1 ± 6.8	4,387 ± 801	186 ± 41	69 ± 12	1.03 ± 0.8
Dissociative group	4,287 ± 542	61.1 ± 6.6	4,428 ± 645	166 ± 21	63 ± 7	1.03 ± 0.5
All groups average	4,272 ± 527	58.8 ± 6.8	4,385 ± 639	178 ± 35	66 ± 10	1.02 ± 0.5

Note: Mean ± SD.

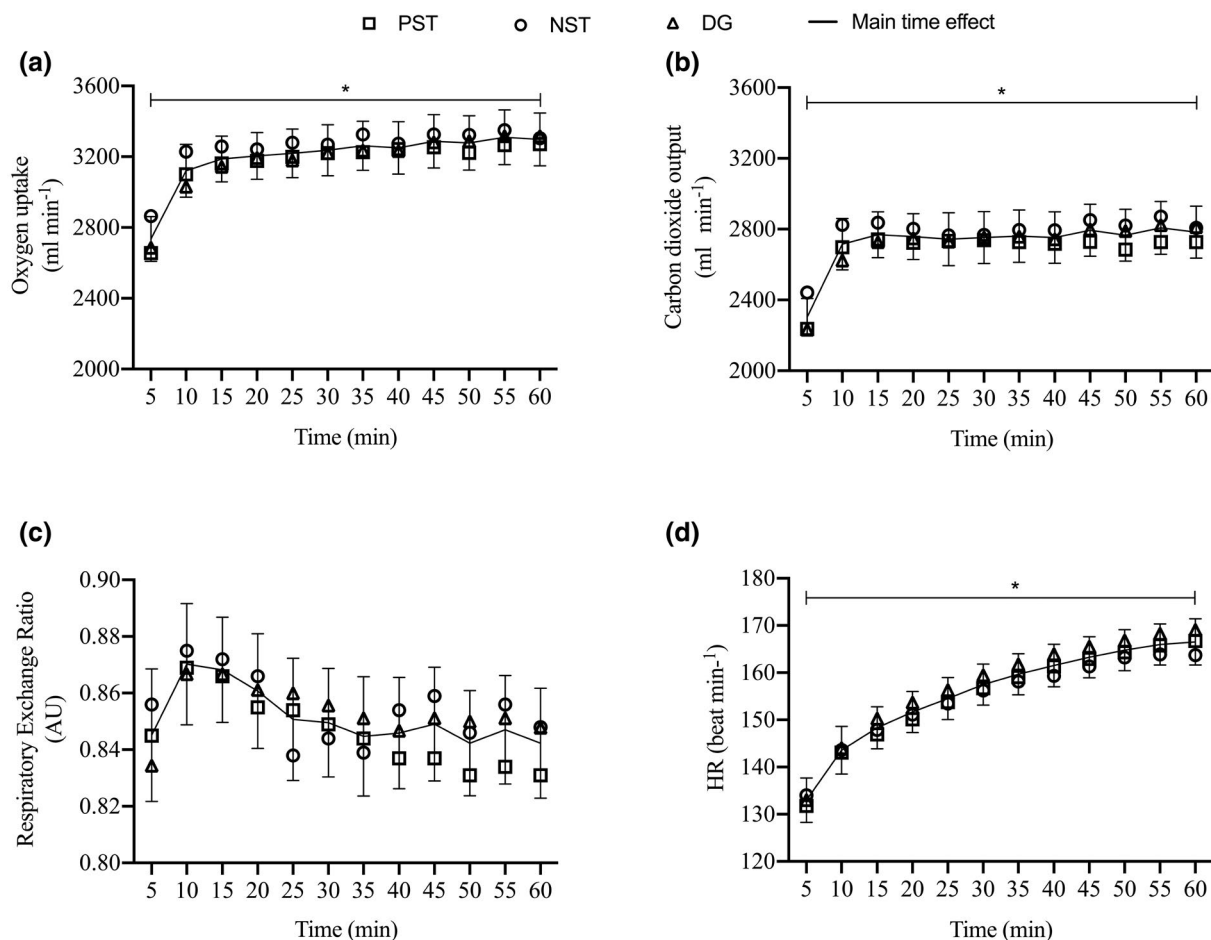


FIGURE 2 Oxygen uptake (a), carbon dioxide output (b), respiratory exchange ratio (c), and heart rate (d) as a function of time during the steady-state running exercise. The lines show the main significant time effect while the square, circle, and triangle display the spread of groups distribution. * indicates $p < .05$ and error bars are 95% CI

changed over time [$\dot{V}O_2$ ($F_{(2,11)} = 4.89$, $p < .001$); $\dot{V}CO_2$ ($F_{(2,11)} = 3.75$, $p < .001$); HR ($F_{(2,11)} = 19.69$, $p < .001$)], a response known as the cardiovascular drift (Coyle & González-Alonso, 2001), no significant interaction or significant main effect of group was found during the 60-min steady-state running exercise.

Figure 3 displays ΔRPE , $\dot{V}_E$, Bf, and V_T over the 60-min steady-state running exercise. There was a significant

main effect of time on $\dot{V}_E$ ($F_{(2,11)} = 9.51$, $p < .001$) and Bf ($F_{(2,11)} = 5.36$, $p < .001$). In addition, the Kruskal–Wallis non-parametric test revealed that ΔRPE significantly increased through time [NST, $H_{(11)} = 51.39$, $p < .001$; PST, $H_{(11)} = 29.33$, $p < .005$; and DG, $H_{(11)} = 34.11$, $p < .001$]. All these outcomes mirror the above-mentioned significant cardiovascular drift. In addition and as displayed on Figure 4, there was a significant main effect of group on $\dot{V}_E$

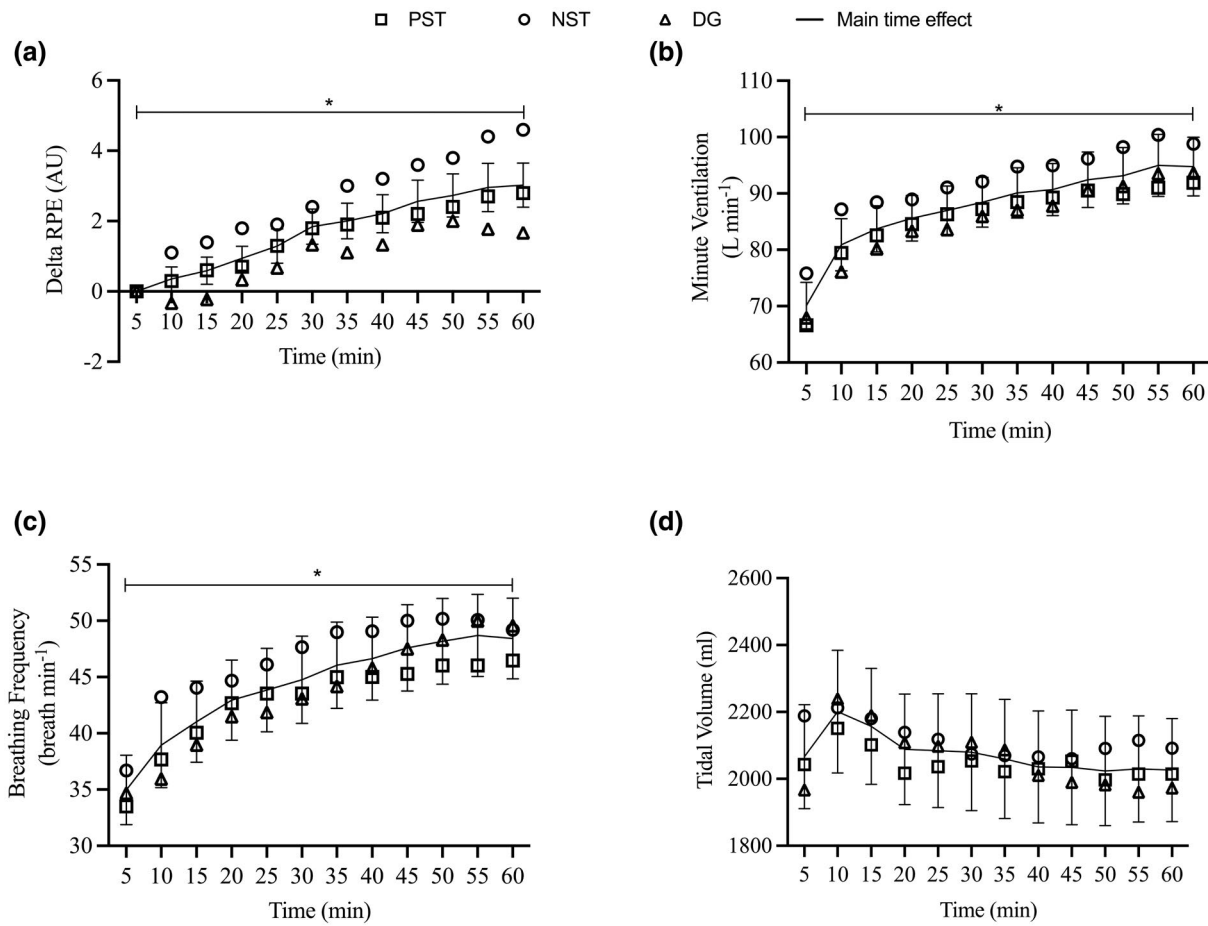


FIGURE 3 Δ rate of perceived exertion (a), minute ventilation (b), breathing frequency (c), and tidal volume (d) as a function of time during the steady-state running exercise. The lines show the main significant time effect while the square, circle, and triangle display the spread of groups distribution. * indicates $p < .05$ error bars are 95% CI

$[F_{(2,11)} = 12.31, p < .001]$, and $Bf [F_{(2,11)} = 5.01; p < .01]$. The *post-hoc* analyses revealed that $\dot{V}_E$ and Bf were significantly higher for NST compared to the two other groups. Figure 4 further displays the normalized cortisol values [normalized from the baseline values] that were log-transformed to approximate a normal distribution. First, there was no significant difference between the first two samples (i.e., after waking and 30-min later) [$\Delta 0.044 \pm 0.014 \mu g \cdot dl^{-1}$; $\Delta 0.052 \pm 0.015 \mu g \cdot dl^{-1}$; $\Delta 0.107 \pm 0.086 \mu g \cdot dl^{-1}$] for NST, PST, and DG, respectively, and between groups [$\Delta PST-NST = 0.003 \pm 0.006 \mu g \cdot dl^{-1}$; $\Delta PST-DG = 0.058 \pm 0.014 \mu g \cdot dl^{-1}$; $\Delta NST-DG = 0.055 \pm 0.021 \mu g \cdot dl^{-1}$]. However, there was a significant main effect of group on normalized cortisol values (i.e., between baseline and post-intervention) [$F_{(2)} = 4.845; p < .03$] and the *post-hoc* analysis showed that NST cortisol level was significantly higher compared to the PST ($p < .005$) and DG ($p < .001$). In addition, the Kruskal-Wallis non-parametric test revealed a difference in the median ΔRPE scores (and, hence, the mean ΔRPE scores) among the three groups ($H_{(2)} = 7.66, p < .03$). Mean delta scores were 2.60 ± 0.19 [95%CI: 1.83–2.98],

1.57 ± 0.13 [95%CI: 1.31–2.22], and 0.96 ± 0.13 [95%CI: 0.69–1.23] for NST, PST, and DG, respectively.

4 | DISCUSSION

To further the interpretation of the study outcomes, it is of primary importance to recall its objectives and experimental design. The intent of this investigation was to examine the effect of the nature of associative self-talk and dissociative focus during prolonged running exercise on physiological variables and the perceived exertion. To achieve this goal, we designed the experiment in such a way that the prolonged running exercise-induced an iso-metabolic stress among all participants (i.e., running at 70% MAS). In doing so, we minimized inter-subject metabolic response variability assuming the physical workload was equivalent for all. In addition, we did recruit well-trained individuals who were active runners from a University Cross-Country running team and local running clubs to avoid the negative impact of physical inactivity on the metabolic

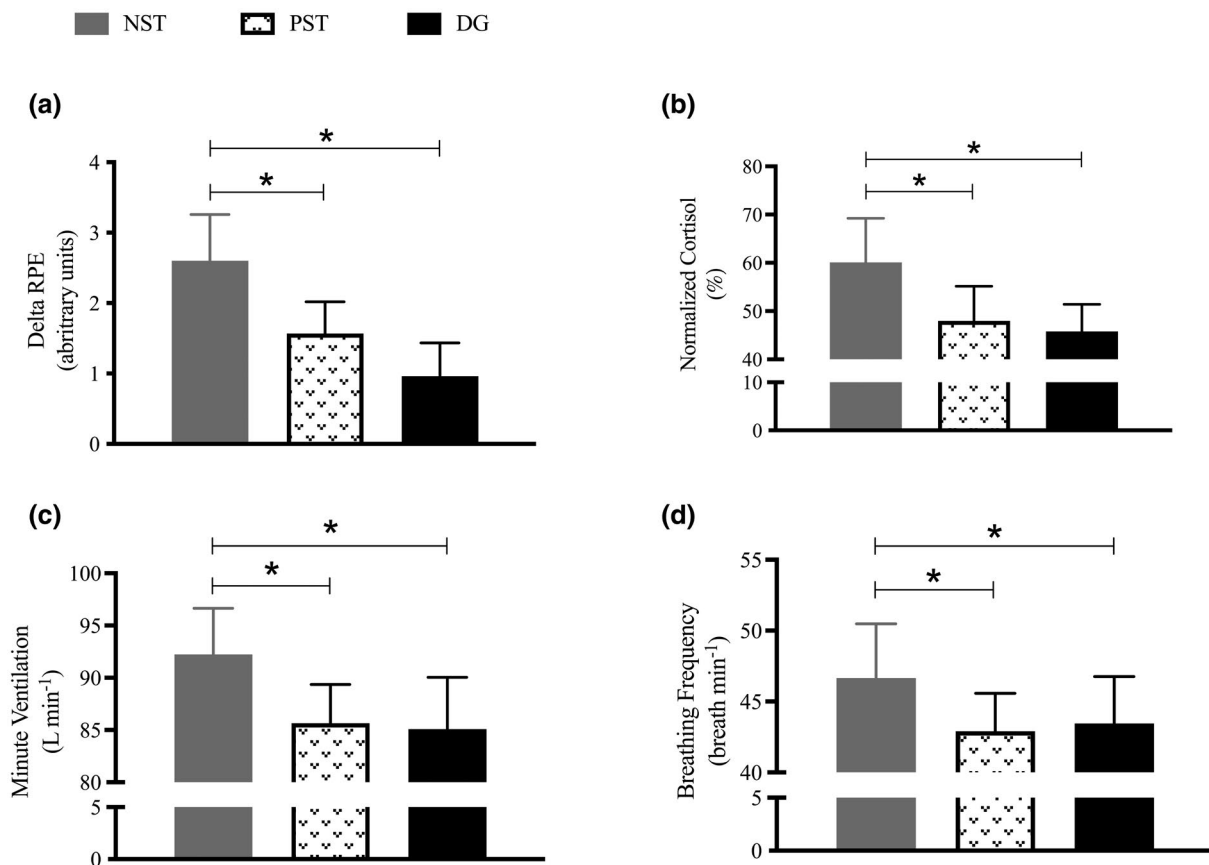


FIGURE 4 The main significant group effect on Δ rate of perceived exertion (a), normalized cortisol (b), minute ventilation (c), and breathing frequency (d) across the steady-state running exercise. The lines show the differences between groups. * indicates $p < .05$ and error bars are ± 1 SD

response to exercise (Booth & Lees, 2006). In this context, we have recorded a psychometric index (RPE; a marker of subjective stress), a humoral stress marker (cortisol), and a cardiorespiratory parameter indicative of the autonomic nervous system activity (breathing frequency) to further our understanding of the complex neural network interplay between the higher cortical areas and the peripheral references during a 60-min steady-state running exercise.

We found a greater HPA response related to increased salivary cortisol in the NST group compared to the other groups (Figure 4B). Moderate-to-high-intensity exercise exceeding 60% of $\dot{V}O_{2max}$ provokes intensity-dependent elevated circulating cortisol levels (Hill et al., 2008). Considering that the relative running intensity was similar between groups (i.e., 70% $\dot{V}O_{2max}$), higher salivary cortisol in the NST group may reflect the additional effect of the nature of self-talk. In that regard, we are extending the work of our predecessors by showing the acute effect of NST on cortisol levels during a 60-min steady-state running exercise.

The association between negative thoughts and cortisol response has been observed in studies on perseverative cognition, such as worry and rumination (Verkuil et al., 2010; Zoccola & Dickerson, 2012). Worry and rumination

are both characterized by repetitive negative thoughts (Watkins, 2008). Perseverative thoughts may lead to prolonged activation of the HPA during problematic goal progress while focusing on unresolved goals in stressful laboratory tasks (Byrd-craven et al., 2010; Zoccola et al., 2008, 2010). Theoretically, perseverative thinking may occur for self-regulation as part of the default response to threat, novelty, and ambiguity (Verkuil et al., 2010); a corresponding emotional defense response in which the HPA activity would be one indicator of the mental stress (Selye, 1956). Nonetheless, this seems not to be exactly the case in the current study since NST was imposed on well-trained runners during mild intensity exercise (i.e., RPE < 15; 70% $\dot{V}O_{2max}$), therefore, not clearly characterizing a context of threat or novelty. Moreover, experienced runners are able to dissociate during 70% MAS since the highest prevalence of associative thoughts occurs at RPE scores between 16 and 20 (Aitchison et al., 2013; Schomer & Connolly, 2002; St Clair Gibson & Foster, 2007). Therefore, we can hypothesize that artificially imposed NST may conflict with the mild running exercise task. Besides, PST and dissociative groups would have matched with the actual physical challenge during the 60-min steady-state running exercise.

In that regard, the self-talk dissonance hypothesis postulates that the attempt to use conscious monitoring with messages that conflict with physiological/emotional state can be detrimental to performance compared to the self-talk that matches the real state (Raalte & Vincent, 2017). Despite the fact that performance was not the goal in the current study, our data add to the dissonance hypothesis by showing that the NST selected by runners during a trivial exercise task heightens HPA and alters cardiorespiratory responses along with greater RPE scores during exercise. Accordingly, future studies may investigate the dissonance effect of PST and NST during iso-metabolic exercises at intensities below 50% MAS (RPE 6–10) and above 70% MAS (RPE 16–20), conditions for which dissociative and associative thoughts are, respectively, more distinctively prevalent than at 70% MAS (RPE 11–15) (Aitchison et al., 2013; Schomer & Connolly, 2002; St Clair Gibson & Foster, 2007).

To date, there have been no studies specifically designed to explore the subcortical neural connections that would elucidate the top-down neurophysiological pathway associating self-talk with physiological responses during exercise. However, we may find parallels in the literature that could inspire future studies about self-talk during exercise. Longe et al. (2010) revealed that the dorsolateral PFC and hippocampal/amygdala complex were positively correlated with an individual's tendency to be self-critical in situations that could be regarded as a personal failure or mistake, which would elicit shame-like negative emotions and threat to self, contrasting with individual's tendency to be self-reassuring and resilient. The amygdala, located within the temporal lobes, stimulates corticotropin-releasing hormone-producing neurons in the hypothalamic paraventricular nucleus and this neuropathway becomes activated in anticipation of potential threat (Herman et al., 2016). Such increased amygdala responses to emotional stimuli have been reported in healthy participants when their cortisol levels were elevated by stress (Henckens et al., 2016). Moreover, a greater daily cortisol level was correlated with increased amygdala connectivity with the hippocampus in responses to fear (Hakamata et al., 2017). Hypothetically, the dissonance between NST and steady-state exercise effort may have triggered the connectivity between the amygdala and the hypothalamic paraventricular nucleus, as in anticipating a potential threat.

The response to threat is also strongly influenced by the extended amygdala-parabrachial circuit (Luskin et al., 2021), a structure that functions as a respiratory pacemaker (Felten et al., 2016). The stimulation of the amygdala produces a rapid increase in respiratory rate unrelated to changes in metabolic demand (Homma & Masaoka, 2008; Masaoka & Homma, 2001). Thus, breathing frequency is not only controlled via the sensory

afferent inputs by metabolic demands but also constantly responds, via the anticipatory feedforward inputs, to changes in emotions, such as sadness, happiness, fear, and anxiety (Masaoka & Homma, 2001). In fact, our data show a mismatch between breathing frequency and the actual physical challenge among the participants randomized to NST. This provides evidence that feedforward inputs alter cardiorespiratory drive during a one-hour steady-state running exercise at 70% $\dot{V}O_{2max}$. Therefore, NST may stimulate the connectivity between the amygdala complex and the respiratory nucleus shooting a physiological response greater than the metabolic demand required. Although technically challenging, the association between NST and subcortical brain activity during exercise could be examined in studies using a cycle ergometer for fMRI (Fontes et al., 2015; Fontes et al., 2020).

The relationship between perceived exertion and central physiological factors such as heart rate and minute ventilation has long been recognized (Borg, 1982; Maddigan et al., 2019; Pandolf, 1978). Moreover, respiratory frequency is strongly correlated with perceived exertion during time trials of different duration (Nicolò et al., 2016). However, the neurobiological basis of perceived exertion is still well debated (Pageaux & Pageaux, 2016). In general, the dispute revolves around whether RPE is dependent on afferent feedback from working muscles (including respiratory muscles) and other interoceptors or is generated by internal neural processes within the brain associated with the central motor command (i.e., activity of pre-motor and motor areas related with voluntary muscle activity) (Pageaux & Pageaux, 2016). In that regard, integrative models postulating that perceived exertion may be a result of neuronal process of sensory signals (i.e., feedback or bottom-up) and psychological factors involving higher cortical areas beyond the primary motor cortex (i.e., feedforward or top-down) has been proposed (Pageaux & Pageaux, 2016; Williamson, 2010). According to the experimental design presented herein, it is not feasible to discriminate whether NST affected RPE directly through internal neuronal processes of the brain and/or indirectly through the amygdala input to the medial parabrachial nucleus. Therefore, it is unknown if the higher RPE is a result of NST (i.e., feedforward effect), increased working respiratory muscles (i.e., feedback effect), or an integration of both. Nevertheless, the feedforward effect associated with the motivational interpretation of induced NST may be ruled out. According to Blanchfield et al. (2014), induced motivational self-talk reduced the perception of effort during a time-to-exhaustion cycling test; however, our results show the opposite: increased RPE in the NST group with similar and lowered responses in the dissociative and PST groups. Differences in exercise tasks may explain the discrepancy found in the present study

as compared to Blanchfield et al. (2014). During a time-trial to exhaustion, athletes truly motivate themselves to reach the peak performance until volitional exhaustion. On the other hand, during a 60-min steady-state running exercise, induced associative self-talk or dissociative focus may represent a cognitive load that would match or not to the exercise effort. As previously suggested, we hypothesize that the dissonance between NST and physical effort may have affected the RPE.

This study suffers from several methodological considerations worthy of discussion. First, we did not incorporate manipulation checks into the experimental protocol out of concern for its influence on the outcome measures. Although participants randomized to the associative groups (i.e., PST or NST) were instructed to repeat the self-talk statements aloud, the extent to which these interventions influenced participants' thoughts is unknown. Furthermore, the induced dissociation chosen herein was intended to be an effective neutral response. In this regard, physiological responses may be distinct in comparison to other techniques for dissociation as high/low tempo music or if the external stimulus is among the subject's affective preferences for music or videos (Ballmann, 2021; Dyrland & Wininger, 2008). Second, the sample size was based on a convenient sample of well-trained individuals and not on an *a priori* statistical power calculation. The recruitment of well-trained individuals might be challenging at times. The participants juggle between training and experimental sessions, often favoring the former to the latter's detriment. Therefore, setting an *a priori* number of participants is not always feasible with this type of population. Third, choosing a one-hour steady-state running exercise at 70% $\dot{V}O_{2max}$ limits the inferences to an unnaturalistic type of exercise. In fact, runners paced themselves during an outdoor one-hour running exercise, contently adjusting speed and effort according to the route's topography, the weather conditions, and the state of fatigue. Fourth, the use and effect of self-talk vary across cultural groups and the language spoken (Raalte & Vincent, 2017). Finally, conclusions might apply to a group of young, well-trained, and fit individuals. Any departure from these conditions must be interpreted with caution.

As a whole, NST increased breathing frequency and cortisol responses during a one-hour steady-state running exercise. Hypothetically, the dissonance between NST and steady-state exercise effort may have triggered the anticipatory response to a potential threat or novelty. This response is similar to the emotional induced cardiorespiratory and cortisol arousal observed before competitive situations, which may prepare the physiological milieu for peak performance (Elias, 1981; Passelergue & Lac, 1999; Viru et al., 2010). However, the assisted NST during a steady-state running exercise also increased the RPE, and

therefore, athletes may reach the maximum perceived exertion (i.e., RPE 20) in a shorter period. In other words, the increase in RPE decreases the time-trial to exhaustion in constant load exercises (Fontes et al., 2010). Thus, future studies may investigate the effect of induced NST on low-to-high constant load time trials in addition to self-paced exercise. In conclusion, the exploratory analysis herein calls for further investigation of the dissonance hypothesis in the interpretation of self-talk content during externally imposed low-to-high exercise intensities. Subcortical targets potentially involved in the dissonance hypothesis, as the connectivity between limbic structures and acute physiological responses, may help identify the neural mediators of the self-talk–physiology relationship.

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AUTHOR CONTRIBUTIONS

Fabien A Basset: Conceptualization; Formal analysis; Investigation; Methodology; Project administration; Supervision; Writing – original draft; Writing – review & editing. **Liam P Kelly:** Conceptualization; Formal analysis; Methodology; Writing – original draft; Writing – review & editing. **Rodrigo Hohl:** Formal analysis; Writing – original draft; Writing – review & editing. **Navin Kaushal:** Conceptualization; Formal analysis; Investigation; Methodology; Project administration; Writing – original draft; Writing – review & editing.

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